

Fine-grained Micro-Expression Generation based on Thin-Plate Spline and Relative AU Constraint

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ABSTRACT

As a typical psychological stress reaction, micro-expression (ME) is usually quickly leaked on a human face and can reveal the true feeling and emotional cognition. Therefore, automatic ME analysis (MEA) has essential applications in safety, clinical and other fields. However, the lack of adequate ME data has severely hindered MEA research. To overcome this dilemma and encouraged by current image generation techniques, this paper proposes a fine-grained ME generation method to enhance ME data in terms of data volume and diversity. Specifically, we first estimate non-linear ME motion using thin-plate spline transformation with a dense motion network. Then, the estimated ME motion transformations, including optical flow and occlusion masks, are sent to the generation network to synthesize the target facial micro-expression. In particular, we obtain the relative action units (AUs) of the source ME to the target face as a constraint to encourage the network to ignore expression-irrelevant movements, thereby generating fine-grained MEs. Through comparative experiments on CASME II, SMIC and SAMM datasets, we demonstrate the effectiveness and superiority of our method. Source code is provided in <https://github.com/MEA-LAB-421/MEGC2022-Generation>

CCS CONCEPTS

• Computing methodologies → Computer vision problems.

KEYWORDS

Facial micro-expression, Micro-expression generation, Image animation, Motion transfer, Generative adversarial network.

ACM Reference Format:

Sirui Zhao, Shukang Yin, Huaying Tang, Rijin Jin, Yifan Xu, Tong Xu, and Enhong Chen. 2022. Fine-grained Micro-Expression Generation based on Thin-Plate Spline and Relative AU Constraint. In *Proceedings of the 30th ACM*

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MM '22, October 10–14, 2022, Lisboa, Portugal

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ACM ISBN 978-1-4503-9203-7/22/10...\$15.00

<https://doi.org/10.1145/3503161.3551597>

International Conference on Multimedia (MM '22), October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3503161.3551597>

1 INTRODUCTION

As a typical psychological stress response, micro-expression (ME) refers to the subtle, spontaneous facial reaction [6, 7], which can reflect a person's psychological state and emotional cognition [2, 14]. Because of this, recognizing MEs can help us understand the true intentions of individuals and has a wide range of applications in national security, psychodiagnosis, business negotiation, and other fields [5, 18, 20, 28]. However, MEs are difficult to perceive by naked eyes due to their short duration (less than 500ms) and weak movement compared to macro-expressions. Although researchers have tried to propose various intelligent automatic ME detection and recognition models in the field of computer vision, their performance are still far from meeting the requirements of practical applications, mainly due to the lack of sufficient well-labeled ME data to train and verify a robust model.

Currently, considering that ME data is difficult to induce and label, many efforts have been made to introduce data augmentation strategies to construct new ME data. These augmentation methods can be divided into two categories, i.e., traditional image augmentation methods and generation-based methods. Traditional image augmentation strategies, such as random image cropping and rotation, are widely used by current MEA methods [12, 13, 19, 27, 29, 30, 36, 37]. Such methods are simple to operate and can quickly generate a large number of homogeneous ME samples, but they usually fail to generate samples with new features.

Meanwhile, the generation-based methods mainly adopt Generative Adversarial Network (GAN) [9] or motion-driven model [23, 24, 35] to generate new ME sequences based on the target face with the ME of driving video, which can effectively enrich the ME data distribution and improve the generalization ability of MER models. For example, encouraged by GANimation for facial expression generation [21], Xie et al. [31] proposed AU-ICGAN model, i.e., AU Intensity Controllable Generative Adversarial Network to generate diverse ME data for MER. Xu et al. [32] proposed FAMGAN, which can generate delicate ME sequences based on fine-grained AU conditions and GANimation framework. Although these conditional GAN (CGAN) based methods can generate new MEs with new identity information, the generated MEs suffer from some defects, such as blurred images, artifacts, and insignificant ME changes. Besides, the training of GAN may be unstable due to the adversarial training procedure. Different from CGAN, motion-driven model usually first

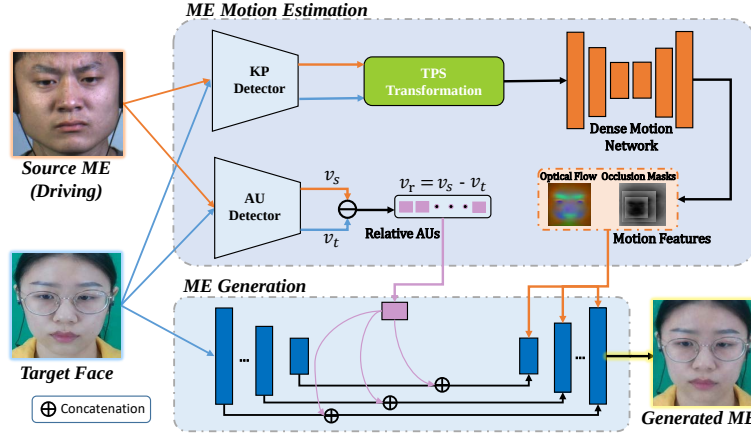


Figure 1: The overall framework of our TPSRAU method mainly includes two key parts, i.e., ME motion estimation based on TPS transformation, and ME generation based on ME motion with relative AU constraint.

models the motion transformation in the driving video, and then manipulates the target face deformation by combining the motion transformation in the generation network to generate the target ME. In the micro-expression generation (MEG) task of MEGC2021, Zhang et al. [34] and Fan et al. [8] obtained surprising results by adopting this strategy. However, these methods still have two shortcomings. First, using the linear changes of key points to model the movement can not adequately represent the nonlinear changes of MEs. Second, expression-irrelevant motions in driving video lead to uncertain motion estimation, which may bring undesirable distortions to unrelated facial regions.

To address the above challenges and encouraged by recent image animation framework [16, 35], we propose a novel fine-grained ME generation method based on Thin-Plate Spline and Relative AU constraint (TPSRAU). Specifically, we first adopted the thin-plate spline transformation with a dense motion network to model the nonlinear ME motion. Then, the estimated ME motion transformations were used to manipulate the generation network to synthesize the target MEs. In particular, we obtain the relative AUs of the source MEs to the target face to constrain the generator network to only transform regions of ME occurs, which are specified by non-zero-valued relative AUs. Finally, we evaluated our method on CASME II, SMIC and SAMM, where our method showed a satisfactory result on all datasets.

2 TECHNICAL FRAMEWORK

2.1 Preliminary and Problem Definition

As mentioned above, the goal of our work is to generate specific MEs on each given template face. Formally, given an input target face image t_i from the template face set T and a driving ME sequence $X = (x_1, x_2, \dots, x_n)$, we target at generating a new ME sequence $Y = (y_1, y_2, \dots, y_n)$, which should contain the identity information of t_i and the ME motion feature from X . Therefore, for each generated image y_i :

$$y_i = \text{Generator}(\text{encoder}(t_i), \text{MT}(x_i)), \quad (1)$$

where *Generator* is the generation network with encoder-decoder structure. t_i represents the target face in the i^{th} template face image, which comes from the driving ME sequence during training. $\text{MT}(\cdot)$ is the ME motion transformation, mapping the ME movements in driving video to the target facial representation. According to Eq. (1), it is easy to understand that estimating reliable ME motion transformation is critical to generating fine-grained MEs.

2.2 Framework Overview

To deal with the problem above and encouraged by the excellent end-to-end unsupervised motion transfer framework for image animation proposed by Zhao et al. [35], we introduce a Fine-grained MEG method based on Thin-Plate Spline and Relative AU Constraint (TPSRAU). It is worth mentioning that there are two key distinctions between our TPSRAU and the framework proposed by Zhao et al. [35]. Firstly, since the background in driving ME video barely changes, we discard the BG Motion Predictor module. Secondly, to ignore expression-independent movements, we adopt relative AUs of the source ME to the target face in the generator network. Intuitively, the overall framework of the refined system is shown in Figure 1, which mainly includes ME Motion Estimation and ME Generation modules.

- **ME Motion Estimation module.** This module aims to estimate the motion transformation between driving ME image x_i and target face T_i , and provides the optical flow and the occlusion masks for target face deformation.
- **ME Generation module.** This module is a regular encoder-decoder based visual generator. In this generator, the encoder is first used to encode the identity representation of the target face. Then, using the obtained optical flow and occlusion masks, the decoder manipulates the target face representation to generate ME.

2.3 ME Motion Estimation based on TPS

Estimating reliable ME Motion transformation is critical for ME generation. Considering that the motion of ME is subtle and nonlinear, we introduce the nonlinear TPS [3] based on the detected

key-points (KP) by the KP Detector [35], to approximate the motion transformations Φ . In general, we can estimate Φ by minimizing the bending energy function J ,

$$J(\Phi) = \sum_{j=1}^2 \iint_{R^2} \left(\frac{\partial^2 \Phi_j}{\partial x^2} \right)^2 + 2 \left(\frac{\partial^2 \Phi_j}{\partial x \partial y} \right)^2 + \left(\frac{\partial^2 \Phi_j}{\partial y^2} \right)^2 dx dy \quad (2)$$

$$s.t. \quad p_i^s = \phi(p_i^t) = \begin{bmatrix} \phi_1(p_i^t) \\ \phi_2(p_i^t) \end{bmatrix}, i = 1, 2, \dots, N$$

where P_i^t denotes the i -th key-point of the target face, P_i^s is the i -th key-point of the ME image, and N is the number of key points.

Afterward, we use the corresponding transformation Φ of each key point to warp the target face t_i , and cascade the warped face images to the dense motion network to generate ME optical flow and multi-resolution occlusion masks for target face deformations.

2.4 ME Generation based on Relative AUs

After obtaining the optical flow and occlusion masks by our ME motion estimation module, we can use them directly to manipulate the target facial representation to generate new MEs by following the previous work [35]. However, due to the uncertainty of motion estimation, this procedure may bring undesirable distortions to irrelevant ME regions. Inspired by [17], we resort to relative AUs to solve this problem. Concretely, we integrate relative AUs into the generation process to create fine-grained ME images.

According to Facial Action Coding System (FACS), facial movements can be coded by a set of Action Units (AUs). Therefore, in a sense, each AU could indicate how strongly a certain facial muscle contracts, and thus acts as an indicator of expression occurs. Based on this fact, an intuition would be that the relative AUs represent the desired changes from one expression to another. Here, the relative AUs are defined as the difference between the target faces and the source ME images. Formally, we calculate the relative AUs V_r through:

$$V_r \triangleq V_s - V_t \quad (3)$$

where V_s denotes the source AU vector of the driving ME image, and V_t denotes the AU vector of the target face. Both of them can be represented by a vector with 17 AU intensity values, each of which indirectly indicates the intensity of movement in each specific face region. According to Eq.(3), the non-zero relative AUs values correspond to the interest AUs of ME occurs and zero-values correspond to unchanged AUs.

After getting the relative AUs vector, we send it together with motion features into the decoder part of the generation network. Concretely, we concatenate relative AUs with warped and masked encoder features as well as features from the previous layer of the decoder. Through this module, our network can better focus on ME-related regions of human faces and perform more delicate manipulations of expressions, thus generating more authentic images.

2.5 Optimization

To optimize the proposed model, we combine two kinds of loss functions, including reconstruction loss L_{rec} [11] and an auxiliary loss L_{warp} [35]. Following previous works [24, 25], a pre-trained VGG-19 [11] network is used for calculating the reconstruction

loss between the driving image x and the generated ME image $\hat{x}$ at multi scale:

$$L_{rec} = \sum_j \sum_i |F_i(x_j) - F_i(\hat{x}_j)| \quad (4)$$

where F_i is the i^{th} layer of the pretrained VGG-19 network, and j is the times for which the image is downsampled. Moreover, we add an auxiliary loss [35] in the generation network to encourage better estimation of motion transformation and to pull closer the warped feature maps of target face t to the feature domain of x :

$$L_{warp} = \sum_i |\Phi(E_i(t)) - E_i(x)| \quad (5)$$

where E_i is the i^{th} layer of the encoder in the generation network and Φ denotes the warp operation based on optical flow. Therefore, the overall optimization objective for the model is:

$$\mathcal{L} = L_{rec} + \lambda \cdot L_{warp} \quad (6)$$

where λ is a hyper-parameter to balance different losses, which is set to 0.9 in our experiments.

3 EXPERIMENTS

In this section, a series of qualitative and quantitative experiments are conducted to evaluate the effectiveness of the proposed generation model, i.e., TPSRAU.

3.1 Datasets

Following the guidance of MEGC2022, three popular spontaneous facial ME datasets are used to train and test our model, including CASME II [33], SAMM [4], and SMIC-HS [15]. For the generation task, three ME videos (i.e., source ME samples) with three different kinds of emotion labels (positive, negative, surprised) are specified from each dataset.

3.2 Implementation Detail

We use Openface Toolkit [1] to detect the 68 facial landmarks and AU intensity vectors, and resize the face region to 256x256. Our TPSRAU was built based on the PyTorch framework, and trained on four GeForce RTX 3090 GPUs with 100 epochs. In TPS, we predict 10 key-point transformations to estimate the motion.

3.3 Evaluation Metrics

Accurately assessing the quality of the generated MEs is difficult because of their short duration and low intensity. Referring to other generation tasks, we evaluated the generated results qualitatively and quantitatively from subjective and objective perspectives.

Objective Metric. Fréchet Inception Distance (FID) [10] is one of the common objective metrics in image generation, but it focuses on general visual features rather than ME features. Therefore, we define the FID-based metric Expression-FID (EFID) metric to evaluate the quality of generated MEs. Different from FID, EFID is calculated based on the fine-tuned Inception-v3 [26] model, which is trained on the composite ME dataset with ME recognition task [22]. Therefore, EFID can better focus on evaluating the ME features of the generated results rather than the general visual features.

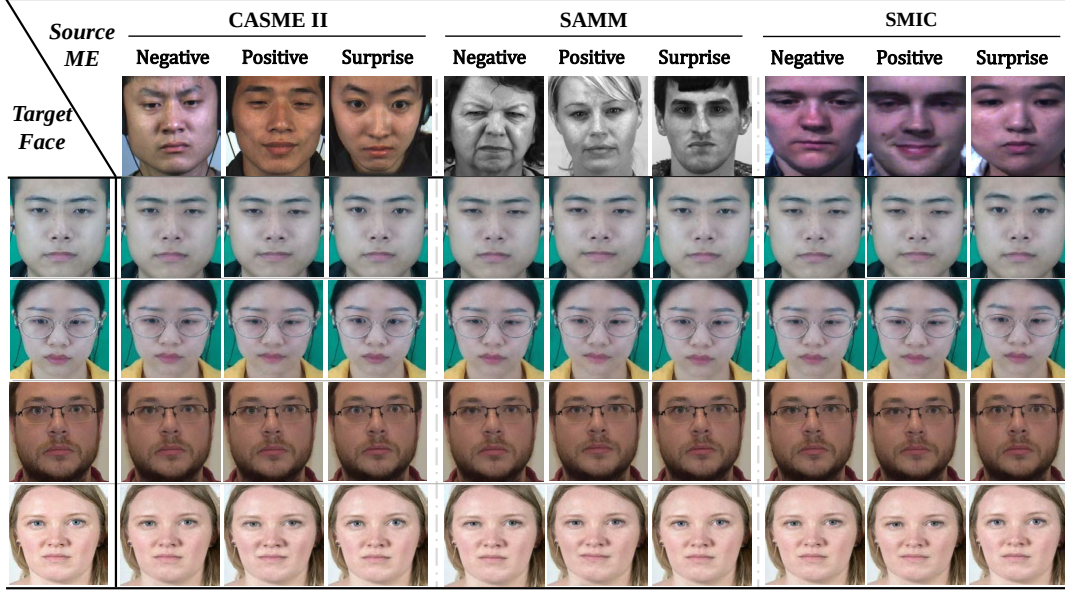


Figure 2: The generated results by our method using four template faces and nine ME sequences, which are specified from CASME II, SAMM and SMIC datasets. It is worth mentioning that due to space constraints, for each generated ME sequence, we only present the apex frame. More details can be viewed from <https://github.com/MEA-LAB-421/MEGC2022-Generation>

Subjective Metric. According to the challenge guideline of MEGC2022¹, three ME experts score the generated results with 0–3 independently from the image quality and AUs, and the final score will be the average of scores provided by three experts.

Table 1: Comparison results of Average EFID on all datasets.

Methods	GANimation ¹	FGFEM ¹	Our (TPSRAU) ²
EFID	116.30	115.67	99.80

¹ CGAN-based Generation method, ² Motion-driven based method.

Table 2: Comparison results of Expert-Score (normalized score) in MEGC2022.

Methods	Expert 1	Expert 2	Expert 3	Overall
# 36	0.61	0.56	0.44	1.61
# 46 (Our)	0.73	0.55	0.58	1.86
# 63	0.74	0.58	0.66	1.98

3.4 Experimental Results

Qualitative results. As with most generation studies, we visually present the generated ME results with apex frames in Figure 2, which contains 36 MEs with 4 different target faces. From Figure 2, it's obvious that the generated MEs can effectively restore the ME motion of the source MEs.

¹https://megc2022.github.io/files/MEGC2022_generation_guideline.pdf

Quantitative results. In this part, we first reproduce two representative CGAN-based expression generation methods as the baselines, including GANimation [21] and FGFEM [17], and compare the generated results on EFID performance, as shown in Table 2. From Table 2, we can easily find that the proposed TPSRAU is significantly outperforms the other CGAN-based methods in EFID performance, which is mainly due to the motion-driven generation framework adopted. In addition, we also report the Expert-Score results of MEGC2022 in Table 2, compared with the other two submitted methods, our method can achieve competitive results.

Ablation Studies. In this part, we perform an relative AU (RAU) ablation experiment to verify the effectiveness of the introduced relative AU constraint. For a clear and dynamic presentation, we present the generated ME results in GIF format with and without RAU module on github². More details are illustrated by <https://mea-lab-421.github.io/MEGC2022-Generation-RAU-ablation>. The ablation results demonstrate that the introduced relative AU constraint can effectively improve the quality of ME generation.

4 CONCLUSION

To generate fine-grained ME, this paper proposed TPSRAU model, a novel ME generation method based on thin-plate spline and relative AU constraint (TPSRAU). We evaluated TPSRAU on the CASME II, SMIC and SAMM, where our method showed satisfactory results.

5 ACKNOWLEDGMENTS

This work was fully supported by the National Natural Science Foundation of China (No.61727809, 62072423).

²<https://mea-lab-421.github.io/MEGC2022-Generation-RAU-ablation>

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